



Fake News Detection

Final Write Up

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Problem Statement:

The rapid development and widespread use of social media and the internet in general have made it much simpler and quicker for information to spread and diffuse. Although the internet makes life easier, it also encourages the spread of false information, which will have a detrimental effect both at large scale with nations to a smaller scope of communities. As a result, numerous studies have been conducted to counteract bogus news. Detecting fake news is usually a classification issue that aims to confirm the accuracy of news information, which might include photographs, videos, and text.

Objective & Approach:

Detect fake news and classify it using the curated datasets we identified and proposed. In our approach, we will focus on machine learning models to classify text-only news statements from these datasets as fake or true.

Why it matters:

- Combat misinformation that can harm public perception and policy.
- Strengthen trust in information sources.
- Fake news threatens truth, trust, and the fabric of society; building detection systems helps safeguard public discourse, improve digital literacy, and ensure people make decisions based on facts—not fabrications

Implementation

The model(s) trained and built in this final implementation were DistilBERT and Naive Bayes (NB). The team implemented a full training pipeline using DistilBERT and Naive Bayes for binary classification with 2 datasets—LIAR and ISOT.

DistilBERT ingests the true and fake datasets from the LIAR and ISOT datasets, merges the title and body of the news text, tokenizes to a fixed maximum length the news text, and fine-tunes the DistilBERT model with subword tokenization of the news texts.

Naive Bayes's implementation follows an adjacent path, albeit with fewer steps. The algorithm relies on clean data, removing elements including punctuation, uppercase formats and bias stopwords. Instead of tokens, the implementation utilizes a Term Frequency, Inverse Document Frequency (TF-IDF) vectorizer to create efficient vector objects that include their rarity weights. Applied to the training and test data, the NB used these objects as a 'vocabulary' to make predictions at lightning speeds.

The modeling approach addresses the core task—binary news fake/real classification. The ISOT dataset was deemed as a better source of data for fake news detection versus the LIAR dataset. After a preliminary model run, we saw that the ISOT data showed better model performance metrics and hence was chosen to be the dataset of choice for

this final implementation. The ISOT data also demonstrated some bias, where the raw ISOT labels conflate “source” and “truthfulness” (e.g., many TRUE samples are Reuters wire or Associated Press stories), so text sanitization was introduced, and deduplication of data is needed to better align the dataset. The LIAR dataset lacked quality of text in the news field, often just 2 to 3 sentences (sometimes a few words) versus multiple paragraphs in the ISOT dataset.

Multiple approaches were used. Beyond the primary DistilBERT and Naive Bayes models, the cumulative effort focused on includes

- Ablations (e.g., sanitization removing publisher markers and stylistic tells; near-duplicate removal before splitting to limit train–test leakage)
- Baseline contrast (e.g., trained sanitized model with sanitized data vs. untrained model with unsanitized data), demonstrating whether performance is driven by shallow text cues vs. deeper semantics
- Real-world samples (a custom dataset of fake news from 2025), testing concept drift and bias in our training sets

Scope changes since the design review were:

The addition of sanitization, near-duplicate control, and data-source-aware analysis was the evolution in scope as the final implementation uncovered these findings. These changes are additive, defensible, and directly motivated by early findings during the design and review phase (e.g., suspiciously high in-distribution accuracy).

Evaluation

Metric design and testing:

The models were evaluated with Accuracy, Precision, Recall, F1, and RMSE (defined on binary predictions; RMSE here equals $\sqrt{\text{error rate}}$), plus confusion matrices.

Baselines and comparisons

Two types of comparisons are particularly useful:

1. Baseline DistilBERT predicting on the original ISOT data test, which showed the baseline model achieving very low scores—evidence that the vanilla Hugging Face model is not trained on the ISOT data

```
Baseline on non-SANITIZED data Distilbert model test:
/tmp/ipython-input-2892752364.py:25: FutureWarning: `tokenizer` argument is deprecated and will be removed in a future version. Please use `tokenizer_class` instead.
  trainer = Trainer(model=model, args=args, tokenizer=tok

=== Baseline on non-SANITIZED test ===
eval_loss: 0.6942
eval_model_preparation_time: 0.0015
eval_accuracy: 0.4905
eval_precision: 0.3423
eval_recall: 0.1137
eval_f1: 0.1706
eval_rmse: 0.7138
eval_runtime: 25.9993
eval_samples_per_second: 298.6230
eval_steps_per_second: 9.3460

Classification report:
              precision    recall  f1-score   support

     REAL         0.52         0.81         0.63         4183
     FAKE         0.34         0.11         0.17         3581

   accuracy              0.49              7764
  macro avg              0.43              7764
weighted avg              0.44              7764

Confusion matrix:
[[3401  782]
 [3174  407]]
```

2. Trained DistilBERT on sanitized data tests, where performance increased significantly and showed that the model was able to predict fake news very well. This is because the sanitized data removed model cues and bias while also indicating the quality of the dataset and the richness of text in the news content. Together, these demonstrate the importance of dataset artifacts into the performance of the models

```
Sanitized-trained on SANITIZED test:
/tmp/ipython-input-2892752364.py:25: FutureWarning: `tokenizer` is deprecated
  trainer = Trainer(model=model, args=args, tokenizer=tokenizer, data_loader=

=== Sanitized-trained Model on SANITIZED test ===
eval_loss: 0.0000
eval_model_preparation_time: 0.0025
eval_accuracy: 1.0000
eval_precision: 1.0000
eval_recall: 1.0000
eval_f1: 1.0000
eval_rmse: 0.0000
eval_runtime: 22.8136
eval_samples_per_second: 340.3230
eval_steps_per_second: 10.6520

Classification report:
              precision    recall  f1-score   support

      REAL         1.00        1.00        1.00        4183
      FAKE         1.00        1.00        1.00        3581

   accuracy                   1.00        7764
  macro avg         1.00        1.00        1.00        7764
 weighted avg         1.00        1.00        1.00        7764

Confusion matrix:
[[4183   0]
 [   0 3581]]
```

In comparison, Naive Bayes tested three baselines:

- 1. ISOT train/test split
- 2. Cross-data evaluation (ISOT train, LIAR test)
- 3. LIAR train/test split
- 4. Real-world fake news

Due to its simple implementation and naturally quick inference, NB was particularly well-equipped to run several tests that provided important insights into our data.

- 1. ISOT Train/Test Split

```

Time to prepare data: 24.03 seconds
Time to test model: 0.09 seconds
Accuracy: 0.937
RMSE: 0.251

```

	precision	recall	f1-score	support
0	0.94	0.94	0.94	4670
1	0.94	0.93	0.93	4310
accuracy			0.94	8980
macro avg	0.94	0.94	0.94	8980
weighted avg	0.94	0.94	0.94	8980

```

Confusion Matrix:
[[4403  267]
 [ 299 4011]]

```

Our initial test included the inference times. In this evaluation, we ran a standard 80/20 split for about 40,000 articles. As shown, most of the effort takes place in preprocessing---data cleaning, vectorization. Highlighting one of NB's strengths, we were able to predict about 8000 articles in about a tenth of a second. Base performance with light ablations resulted in highly positive performance across evaluation metrics, including balanced false positives and false negatives.

2. Cross-data Evaluation

```

cross_data_preds= model.predict(dfv_vectors)
print(f"Accuracy: {accuracy_score(dfv_y, cross_data_preds)}")
print(classification_report(dfv_y, cross_data_preds))

```

```

Accuracy: 0.5771028037383178

```

	precision	recall	f1-score	support
0	0.72	0.61	0.66	864
1	0.39	0.51	0.44	420
accuracy			0.58	1284
macro avg	0.55	0.56	0.55	1284
weighted avg	0.61	0.58	0.59	1284

We then tested the trained model on samples from the LIAR dataset. We discovered much lower scores, although FP and FN were still relatively balanced. In this test, we expected to see an accuracy drop <10%, but saw closer to 30% This led us into an investigation of our two datasets, prompting our third test strictly using LIAR. Here, we learned that these datasets had distinct differences in context size (about 100x less on LIAR)

3. LIAR Train/Test Split

```

liar_preds= model.predict(df_test_vectors)
print(f"Accuracy: {accuracy_score(df_test_y, liar_preds)}")
print(classification_report(df_test_y, liar_preds))

```

```

Accuracy: 0.6791277258566978

```

	precision	recall	f1-score	support
0	0.68	0.98	0.80	864
1	0.59	0.06	0.11	420
accuracy			0.68	1284
macro avg	0.64	0.52	0.46	1284
weighted avg	0.65	0.68	0.58	1284

Using similar parameters, we loaded the LIAR dataset into our NB model. Again, we discovered low scores; the model particularly struggled in recall. We hypothesized an issue with the amount of context and chose to experiment on real-world articles that matched the size of those in ISOT.

4. Real-world Fake News

```

rw_df= pd.read_csv('/content/drive/MyDrive/colab_notebooks/datasets/504_G6/504_RW_test.csv')
rw_test= rw_df['text']
rw_test= rw_test.apply(clean_text)
rw_test_vectors= vectorizer.transform(rw_test)

rw_pred= model.predict(rw_test_vectors)
print(rw_pred)

```

```
[1 0 1 0 0 0 1 1 1 1 0 1 0 0 1 0 0]
```

We pulled together 17 high-bias articles from notoriously untrustworthy publications to test against ISOT. These articles varied in their bias, with 11 coming from far-right sites and six from far-left sites. You can view the dataset in the references and sources below.

With articles ranging from 500-1500 words, we expected the model to have only slight inaccuracies, potentially due to concept drift. We were hoping to see only two to three misclassifications, but received eight. This led to discussions on the effects of concept drift and the quality of ISOT used as a training set.

Findings

- On original, unsanitized test data, the untrained DistilBERT model attains 49% accuracy—consistent with model generalization for data it has not seen before.
- On sanitized test data (publisher markers removed, caps/punctuation normalized), scores were greatly improved, indicating prior reliance on superficial cues and bias in data.

- The gap between original vs. sanitized evaluations quantifies how much performance hinged on data artifacts and the ability to tune and sanitize data.
- Naive Bayes excelled in inference time, predicting 8000 articles in 1/10 second.
- NB also performed surprisingly well on ISOT train/test, with 94% error scores across the board.
- NB may be particularly sensitive to concept drift and its vectorized vocabulary.

Discussion

What worked:

- End-to-end data engineering quality: we determined that among the 2 datasets, the ISOT dataset worked best and had the best metrics suited for our problem due to the verbose nature of the news text and the source from which it originated. The binary class distribution of the label was very even, allowing for better model performance and reducing bias on a single class.
- Using a breadth of performance metrics like precision, recall, F1 alongside accuracy, RMSE, confusion matrices: this helped evaluate the model performance on the dataset used.
- Deduplication and sanitization were the right levers for this dataset and showed a strong correlation to text-classification pitfalls and bias.
- Model trained successfully on ISOT dataset and was able to predict fake news with high accuracy.
- NB's straightforward implementation made it easy to set up and understand. Interpretation was quick to see.
- NB's quick preprocessing and inference allowed us to run several tests, leading to our conclusions regarding the LIAR dataset. This tactic can be used practically to run many tests in a short amount of time before applying insights to more complex models
- With enough data, NB performs surprisingly well. This can be further augmented with fine-tuned datasets. We can use this benchmark to confirm data quality.

What didn't work or lessons learned.

- In-distribution inflation of the ISOT's data construction makes it easy to overestimate generalization if you do a random split without sanitization. The data was better sanitized
- Limits of "style detection": low scores or bias can be explained by publisher tokens (e.g., "Reuters"), editorial structure, and tone markers (ALL-CAPS,

excessive punctuation). Source bias (the #1 reason). ISOT's *True* articles largely come from Reuters, while *Fake* articles come from a small set of fringe sites. Models learn publisher stylistic fingerprints (e.g., "(Reuters) —", dateline formats like "WASHINGTON (Reuters) — ...", editor bylines, and AP-style punctuation) rather than truthfulness.

- Generalization remains an open challenge: even after sanitization, the model has not yet been validated on out-of-domain datasets (e.g., LIAR, PolitiFact & GossipCop) or new sources of fake news data, which would be the gold standard to demonstrate robustness.
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- The dataset contains highly similar articles (wire updates, reprints, and copy-paste variants). Even after removing exact duplicates, near-duplicates can land in both training and testing after a random split, inflating test accuracy.
- NB really struggles when not provided enough context. Users must ensure their implementation has a deep enough library to overcome nuance and sarcasm
- NB is sensitive to its training data. The model builds its vocabulary using what it's given to provide probabilistic interpretations. If tested on information outside of its vocabulary, it will struggle with predictions
- The model's static vocabulary also makes it sensitive to concept drift. For example, NB trained on 'Pandemic' focused articles may miss out on important words from 'Israel-Palestine' focused articles.

Next steps or future considerations:

- Source-held-out testing (train on some other news outlets, test on unseen news outlets) to measure source-agnostic performance.
- Temporal splits (train earlier, test later) to assess drift and topic shift. Cross-dataset evaluation to quantify and reduce dataset-specific bias.
- De-duplicate more aggressively. Use sentence/paragraph embeddings & cosine similarity (or MinHash) to remove near-duplicates prior to splitting.
- Train with source-balanced data (multiple reputable outlets for REAL; multiple rumor/satire sites for FAKE) and hold out entire sources for testing.
- Add adversarial and contrastive negatives (reliable-source articles with sensational headlines or cleanly written fakes) to break the style shortcut.
- Consider evidence-based setups (claim & retrieved evidence) where the model must weigh external facts, not just writing style.

Deployment Considerations

Deployment of the DistilBERT model in the real world needs to consider:

1. Risk controls & governance
 - Calibrate probabilities: set conservative thresholds to avoid over-blocking. What action follows a “FAKE” prediction—flag, downrank, block, or route to review? Different actions imply different precision & recall targets.
 - If false positives are costly (blocking real news), optimize **high precision** for the FAKE class (e.g., $P \geq 0.95$).
 - Human-in-the-loop review for edge cases and appeals.
 - Monitoring for data drift, publisher mix changes, and adversarial style of testing.
2. Data & labeling strategy
 - Expand beyond ISOT with multi-source, balanced real & fake examples.
 - Include counter-examples (well-written fakes, sensational but real articles) to break style shortcuts.
3. Methodology evolution
 - Add adversarial training against publisher cues. Enforce sanitization in production preprocessing (strip boilerplate, normalize punctuation/casing).
4. Privacy, compliance, and transparency
 - Document failure modes and publish model characteristics and datasets for transparency.
 - Ensure compliance with content policies and legal constraints when flagging & acting on outputs.
5. Architectural considerations – GPU for model inference and scalability of use

On the other hand, Naive Bayes implementation should consider:

1. **Domain-focused build:** if implementing NB for a specific industry, use training data that represents that sector. The more its vocabulary matches the content in question, the better it will perform.

2. **Data drives quality:** knowing where the information comes from can help guide improvements. For example, when testing ISOT on real-world articles, it predicted conservative content as likely fake and liberal content as likely true. If ISOT's fake articles came strictly from right-wing sites, that might explain why it leans this way.
3. **Appropriate for quick, non-critical problems:** Naïve Bayes is commonly used to detect spam emails, where tones, nuance and sarcasm typically don't exist. Use NB mainly to gain quick insights on text but not to make important decisions. Thus, when solving for fake news, where a misclassification can spread potentially harmful information, opt for a more complex model that can deeply understand context.

Project Team Work Plan

Subject Area	Task	Lead	Secondary	Comments
Dataset 1: LIAR	Exploration	Dheepa	Venu	copy data files and load
	Cleansing, Preparation	William	Sunitha	
Model 1: DistilBert*	Design, Build	Sunitha	Dheepa	
	Train, initial test	Venu	William	
	Validation, optimization	Dheepa	Venu	
Dataset 2: ISOT	Exploration	William	Sunitha	
	Cleansing, Preparation	Sunitha	Dheepa	
Model 2: Naive Bayes	Design, Build	William	Sunitha	
	Train, initial test	Dheepa	Venu	
	Validation, optimization	William	Sunitha	

Final Deliverables	Write Up	Sunitha	Dheepa	
	Presentation	Venu	William	

References and Sources

SMU Scholar. (2018). [Fake News Detection: A Deep Learning Approach.](#)

LIAR Dataset

William Yang Wang. (2017). [“Liar, Liar Pants on Fire”: A New Benchmark Dataset for Fake News Detection.](#)

[Download for LIAR Dataset TSVs](#)

ISOT Dataset

Ahmed H, Traore I, Saad S. (2018). [Detecting opinion spams and fake news using text classification.](#)

Ahmed H, Traore I, Saad S. (2017). [Detection of Online Fake News Using N-Gram Analysis and Machine. Learning Techniques.](#)

[Download for ISOT Dataset CSVs](#)

[Our Naive Bayes Implementation \(Google Colab\)](#)

[Our DistilBERT Implementation \(Google Colab\)](#)

[Real-world Fake News Dataset](#)